Abdullah Nayem Wasi Emran

Lecturer, Brac University (CSE)
BSc. Graduate of 2025,
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Research Interest

AI/ML for Healthcare, Computational Biology & Bioinformatics, Causal & Fair ML, Health & Medical Informatics

Education

- **Bangladesh University of Engineering and Technology (BUET)** — Feb 2020 - Mar 2025
  BSc. (Engg.) in Computer Science and Engineering — Dhaka, Bangladesh
  - **Honors:** Dean's list award and university merit scholarship recipient in four out of eight semesters
  - **Relevant Coursework:** Machine Learning, Computer Security, Introduction to Bioinformatics, Computer Networks, Data Structures and Algorithms, Database, Linear Algebra, Software Engineering

Teaching Experience

- **BRAC University** — June 2025 - Present
  Lecturer, Department of Computer Science and Engineering — Dhaka, Bangladesh
  - **Teaching:** Discrete Mathematics, Compiler Design, Computer Networks, Bioinformatics

Publications

- **Abdullah Nayem Wasi Emran**, Majisha Jahan Disha, Rezwana Reaz, "Fusing Exploration and Exploitation: Hybrid Fireworks–Whale Optimization for Multi-Objective Independent Task Scheduling in Cloud Environments," *Accepted at Cluster Computing*, 2026.
- **Abdullah Nayem Wasi Emran**, A. B. M. Alim Al Islam, "Domain Adaptive Uplift Modeling across Heterogeneous Mental Health Cohorts," *iScience*, 2026. DOI: 10.1016/j.isci.2026.116027.
- **Abdullah Nayem Wasi Emran**, Tanveer Rahman, "SCOPES: Stability-Aware Cross-Platform Feature Selection for Matched TCGA Gene Expression and RNA-Seq Data," AAAI 2026 Workshop on AI to Accelerate Science and Engineering (AI2ASE), 2026. Camera-ready PDF.
- **Abdullah Nayem Wasi Emran**, A. B. M. Alim Al Islam, "Mental Health Across Contexts: A Cross-Dataset Study Covering Medical Students, Quarantined Individuals, and Psychiatric-Disordered Subjects," *Humanities & Social Sciences Communications*, 2025. DOI: 10.1057/s41599-025-05053-x.

Research Experience

- **Efficient RNA–Protein Interaction Prediction with Foundation-Model Embeddings** — July 2025 – Present
  Supervised by Dr. Mohammad Saifur Rahman
  (Bioinformatics, Deep Learning)
  - Developing RNA–protein interaction prediction models using fixed foundation-model embeddings for RNAs and proteins, with experiments on standard RPI benchmarks and external cold-start settings.
  - Implemented lightweight embedding-based and graph-based architectures, including PairMLP variants and prototype-conditioned graph models, under edge-level, RNA-cold, and protein-cold evaluation protocols.
- **Domain-Adaptive Uplift Modeling across Heterogeneous Mental-Health Cohorts** — Apr 2025 – Jan 2026
  Supervised by Dr. A. B. M. Alim Al Islam
  (Causal Inference, Machine Learning)
  - Harmonizing heterogeneous cohort data (students, quarantined, clinical) into a comparable schema.

- Training a domain-adversarial uplift model with fairness-aware regularization and evaluating with leave-one-cohort-out AUUC.
- Studying cohort-specific calibration to handle inverted or unstable treatment effects.

- **Hybrid Fireworks–Whale Optimization for Multi-Objective Cloud Task Scheduling** Nov 2023 – Mar 2025
  Supervised by Dr. Rezwana Reaz
  (Cloud Computing, Optimization)
  - Designing hybrid meta-heuristics that fuse exploration (FWA) with exploitation (WOA) for cloud task scheduling.
  - Benchmarking methods and analyzing trade-offs in makespan, monetary cost, and CPU/RAM/bandwidth usage.

- **Mental Health Across Contexts: A Cross-Dataset Study** Jul 2024 – Jan 2025
  Supervised by Dr. A. B. M. Alim Al Islam
  (Data Mining, Network Analysis)
  - Analyzing mental-health outcomes across medical students, quarantined individuals, and psychiatric-patient cohorts.
  - Applying statistical modeling, network analysis, and causal methods to uncover robust cross-dataset trends.

Academic Projects

- **SCOPES: Cross-Platform Gene Signature Selection**
  Bioinformatics Course Project [repo]
  - Built a leak-free, multi-objective pipeline (AUC, Kuncheva stability, MMD) to select platform-invariant gene subsets on Agilent that transfer to RNA-Seq without retraining.
  - On TCGA-BRCA (505 tumor / 25 normal), achieved $\text{AUC}_{\text{Agilent}}=0.69$, $\text{AUC}_{\text{RNA-Seq}}=0.61$ ($\Delta\text{AUC}=-0.08$) with a 1-gene model; contrasted with a 30-gene model (near-perfect source AUC, poor transfer).
  - **Tools & Technology:** Python (scikit-learn, pymoo), NSGA-II, MAD filter, AUC/MMD/Kuncheva

- **Lightweight Deep Learning Models for Histopathological Cancer Cell Classification**
  Machine Learning Course Project [repo]
  - Designed and implemented lightweight models using EfficientNet-B0 to classify histopathological cancer cells.
  - Integrated attention mechanisms like SEBlock, MSFF, and CBAM to enhance feature recalibration and spatial learning.
  - Used transfer learning (PathMNIST) and augmentations, achieving 98% accuracy on CRC-VAL-HE-7K dataset.
  - **Tools & Technology:** Python (PyTorch), SEBlock, CBAM, EfficientNet-B0, Matplotlib

- **Ray-Tracing Pipeline**
  Computer Graphics Sessional [repo]
  - Developed a raster-based graphics pipeline and implemented ray tracing using C++ and OpenGL.
  - Applied Phong Lighting model (ambient, diffuse, specular, recursive reflection) to render realistic shapes.
  - **Tools & Technology:** C++, OpenGL

- **RecipeShare**
  Software Development Project [repo]
  - Developed a recipe-sharing web application with features for upload, editing, blog posts, and meal planning.
  - Implemented recipe generation from images of ingredients using backend processing.
  - **Tools & Technology:** Next.js, Django, PostgreSQL

- **Building a Subset C Compiler**
  Compiler Sessional [repo]
  - Developed a compiler from scratch in C++ using Bison and Flex.
  - Implemented symbol table, lexical analyzer, semantic analyzer, and three-address code generation.
  - **Tools & Technology:** C++, Bison, Flex

- **BookExchange**
  ISD Course Project [repo]
  - Built a platform for users to sell, borrow, rent, or exchange books with reviews and search functionality.
  - **Tools & Technology:** Next.js, Django, PostgreSQL

- **Network Simulations & TCP Adaptive Reno**
  Networking Sessional [repo]
  - Implemented socket programming in Java for inter-system communication.
  - Simulated network architectures with NS-3 and explored TCP Adaptive Reno congestion control.
  - **Tools & Technology:** Java, NS-3

Awards and Honors

- **Dean's List Award and University Merit Scholarship:** Academic excellence at BUET 2022,2023

Skills

- **Programming Languages:** Python, C/C++, Java, JavaScript, Bash, MySQL, HTML/CSS
- **Tools & Frameworks:** NS-3, Next.js, Django, Oracle DBMS, Git, LaTeX, Autopsy
- **Libraries:** OpenGL, Scikit-learn, Pandas, Matplotlib, Seaborn
- **Software:** Navicat, Wireshark, MS Office, Adobe Photoshop

References

- **Dr. Mohammad Saifur Rahman** Email: **mrahman@cse.buet.ac.bd**
  Professor, Department of CSE, BUET [Graduate Thesis Supervisor]
- **Dr. A. B. M. Alim Al Islam** Email: **alim_razi@cse.buet.ac.bd**
  Professor, Department of CSE, BUET [Research Supervisor]
- **Dr. Rezwana Reaz** Email: **rimpi@cse.buet.ac.bd**
  Associate Professor, Department of CSE, BUET [Undergraduate Thesis Supervisor]